

Ian Edmundson

Data Scientist

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Personal Summary

Data scientist with 3+ years building analytics infrastructure and governance programs for federal operations. Developed forecasting systems, data quality standards, and reporting platforms serving 30+ stakeholders across technical and executive audiences. Recognized with an NIH Director's Award for analytics work that protected critical infrastructure during a high-demand crisis event. Passionate about using data to align cross-functional teams around shared public service missions.

Technical Capabilities

Predictive Analytics: Built forecasting and anomaly detection systems to anticipate operational issues (Python, TensorFlow, XGBoost, Sklearn)

Cloud Infrastructure: Deployed scalable data platforms and ML pipelines (Azure ML, Databricks, Docker, GitHub Actions)

Data Engineering: Designed pipelines and APIs for reliable data flow (Python, SQL, FastAPI, pandas, Osisoft PI System)

Visualization & Reporting: Created dashboards translating technical data into actionable insights (Plotly, Dash)

Experience

Mentor | Great Learning

Feb. 2026 – Present

Mentoring adult learners in Johns Hopkins University's continuing education AI program

- Deliver weekly office hours to guide learners through hands-on technical coursework

Technical Volunteer | University of Washington Botanic Gardens

Oct. 2025 – Present

Contributing to frontend and database infrastructure for seed conservation programs

- Modernizing a legacy .NET/C# web application for RareCare, a UW seed conservation collections project, migrating codebase to GitHub, refactoring business logic, improving frontend functionality, and extending database capabilities with new tables and schema improvements

Data Scientist (Federal, GS-11) | National Institutes of Health (NIH)

Jan. 2022 – July 2025

Drove data analytics (AI, ML, Forecasting), infrastructure development, and reporting for NIH's 300+ acre research campus facilities operations

Data Governance & Program Development

- Established Databricks analytics platform with Delta Live Tables processing 30M+ daily sensor readings across 35,000+ monitoring points with full data lineage tracking, delivering anomaly detection, predictive forecasting, and automated reporting to 30+ stakeholders that supported \$2.2M in annual energy savings
- Partnered with subject matter experts to establish data quality standards, including expected operating ranges, sampling rates, and timeout thresholds, enabling automated identification of anomalous and stale sensor readings across 20,000+ monitoring points and ensuring data integrity for downstream analytics and reporting
- Administered Azure resource governance, monitoring ~\$120K in annual cloud infrastructure spend and managing user access controls and linked services to ensure secure, compliant infrastructure for analytics applications
- Created standard operating procedures for analytics program infrastructure, establishing technical protocols and workflow documentation that reduced onboarding time for new team members

Data Infrastructure & Platform Engineering

- Collaborated with IT and facilities teams to migrate 6 years of sensor data (35,000+ monitoring points) to cloud infrastructure, achieving 83% faster transfer speeds by developing custom open-source solution instead of using vendor tools, enabling scalable analytics previously impossible on legacy systems
- Migrated 15+ ML models from siloed CSV-based experimentation to a cloud-native MLOps pipeline with managed feature tables, cutting model deployment from weeks to days and accelerating time-to-insight for operational decision-making across a team of 5
- Implemented CI/CD pipelines for 10+ analytics applications, cutting deployment cycles by 50% and eliminating manual release processes across dev, test, and production environments
- Refactored legacy analytics applications into documented, scalable APIs, enabling reliable data integration across systems and simplifying ongoing maintenance of previously unmaintained codebases

Predictive Analytics & Operational Decision Support

- Engineered 96-hour cooling demand forecasting system that was 50% more accurate than the legacy system, enabling proactive resource scheduling and precise equipment sequencing that directly prevented failures during peak demand, supporting safe operation of NIH's 62,400-ton cooling infrastructure
- Created an AI search system enabling instant retrieval of building codes and safety policies from 1,000+ page compliance manuals, reducing query time from manual searches to seconds and ensuring staff could access regulatory standards at the point of decision
- Developed anomaly detection models that reduced false positive maintenance alerts by 93% across 20,000+ monitored points, enabling engineers to identify and prioritize genuine equipment risk, reduce unplanned downtime, and allocate resources to highest-need failures across critical utility systems
- Built a graph-based AI system that extracted entities and relationships from 500+ operator text logs to construct an interactive knowledge graph, enabling leadership to visually explore connections between equipment failures and operational patterns to support root cause analysis and executive reporting

Data Analyst | Contractor to NIH

Oct. 2021 – Dec. 2021

Delivered analytical support for Central Utility Plant optimization and troubleshooting

- Analyzed engineering and historical data to troubleshoot Central Utility Plant faults and support optimization program refinements

Facilities Security Engineer | Cape Fox Corporation (Contractor to NIH)

May. 2021 – Sept. 2021

Delivered mechanical engineering and monitoring system support for central utilities infrastructure

- Compiled monitoring point documentation and sensor configuration data across CUP infrastructure
- Maintained uninterrupted data collection across critical facilities infrastructure by troubleshooting OSIsoft PI historian failures, preserving data integrity for facility monitoring operations

EDUCATION

University of Maryland

May. 2021

Bachelor of Science, Mechanical Engineering

Awards

NIH Director's Award

2024

Recognized for developing ML-based 96-hour forecasting and optimization system that enabled safe operation of NIH's 62,400-ton cooling infrastructure during 2023 El Niño heatwave, when demand approached maximum capacity and required precise chiller sequencing

Certifications

- Applied Data Science Program: Leveraging AI for Effective Decision-Making (MIT Professional Education)
- NIH Training Center Emerging Talent Program (NIH)
- OLAO Lean Six Sigma Green Belt Training Course (NIH)
- Building Knowledge Graphs with LLMs (Neo4j)